

TrES-3b

R-2666

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_110927 · created 2026-08-30T00:50:14+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2455831.7073 +/- 0.00097 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.196 +/- 0.056
Transit depth (Rp/Rs)^2	3.84 +/- 2.21 [%]
Orbital Inclination (inc)	84.55 +/- 2.87
Airmass coefficient 1 (a1)	0.81 +/- 0.12
Airmass coefficient 2 (a2)	0.095 +/- 0.038
Scatter in the residuals of the lightcurve fit is	13.39 %
Best Comparison Star	#3 - [246, 398]
Optimal Aperture	4.03
Optimal Annulus	5.72
Transit Duration (day)	0.07 +/- 0.019

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.196 ± 0.056 vs 0.16309 (deviation 0.59σ)
Duration fit vs published	0.07 d vs 0.0567 d (deviation 0.7σ)
Mid-time O–C	1.4 min
Depth SNR	3.5
Residual scatter	13.39 %

Light curve

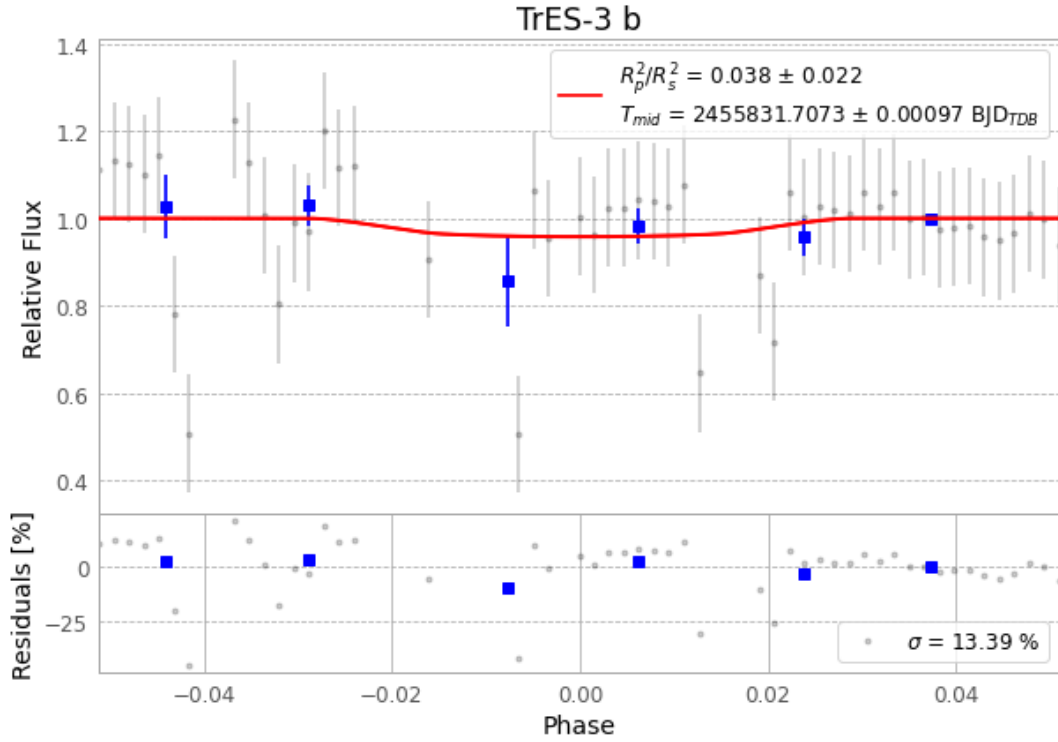


Figure 1 — FinalLightCurve_TrES-3 b_2011-09-26.png (SHA-256 c872cf0db7cb1a68...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [315, 314], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8c6dc71776b00a80...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

63 FITS frames (41 MB), TRES-3110927032113.FITS → TRES-3110927063312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-110927.log (32b78dd25c8f...), FinalLightCurve_TrES-3 b_2011-09-26.png (c872cf0db7cb...), FinalLightCurve_TrES-3 b_2011-09-26.csv (220a9a8aea43...)

Compute resources

Wall time 164.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-30 00:50Z.